Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/25/2006
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Project Title:	The Influence of Rotational Cues on Human Tilt and Translation Responses		
Division Name:	Human Research		
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Comments:	In August 2005, the PI was promoted Assistant-Professor in the Dept of Otology and Laryngology at Harvard Medical School.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
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Contact Monitor:		**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: PI is now Dan Merfeld (per P. Denkins/JSC), so closing this task (jvp 4/07) NOTE: Funding reinstated to 6/1/2008, per J. Cohen (HQ, 8/2006)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Merfeld, Daniel (Massachusetts Eye and Ear Infirmary)		
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Task Description:	Sensory systems often provide ambiguous information. For example, otolith organs measure gravito-inertial force, the sum of gravitational force and inertial force due to linear acceleration. According to Einstein's Equivalence Principle, no set of linear accelerometers alone can distinguish gravitational force (which changes with head orientation during a head tilt) from inertial force (which changes with linear acceleration of the head). Therefore, the Central Nervous System (CNS) must use other sensory cues to distinguish tilt from translation. For example, the CNS can use rotational cues provided by the semicircular canals and vision. To investigate the influence of rotational cues on the neural processing of tilt and translation in humans, we propose to develop a countermeasure to adapt and/or pre-adapt astronaut canal-otolith interactions to any gravitational environment (Earth, Mars, …) by providing correlated sensory cues between the canals and the otoliths. The countermeasure we propose to develop, often referred to as Artificial Gravity, consists of centrifugation combined with intermittent motion. We will measure behavioral (eye movements such as vestibulo-ocular reflexes) as well as psychophysical responses (orientation estimation) during tilt alone, translation alone or combined tilt-translation motion in either a 1-G or hypergravic environment. Fourier analysis and modeling tools will also be used to investigate the neural processing between rotational and gravito-inertial cues. This Task addresses the following specific aims (SA) using human subjects: SA I. Using psychophysics and eye movement measurements, investigate how humans utilize semicircular canal cues to help estimate the relative orientation of gravity (monitored through perceptual tilt responses) as well as the estimate of linear acceleration (monitored through translational VOR responses). This specific aim addresses Critical Question 9.01 since SA 1 will provide baseline data for humans to evaluate if Artificial Gravity combined with intermittent motion is a valid countermeasure (SA 2). We propose a ground study to measure human responses (perceptual tilt and eye movements) during combined canal-otolith stimulation on Earth, in a 1-G environment. Our goal is to provide a perceptual measure of tilt with simultaneous eye movement recording to comprehensively investigate how the human Central Nervous System distinguishes tilt from translation. Subjects will be tested using protocols in which they passively a) tilt, b) translate, or c) tilt & translate simultaneously. SA II. Using psychophysics and eye movement measurements, investigate how humans utilize semicircular canal cues to help estimate the relative orientation of gravity (monitored through perceptual tilt responses) as well as the estimate of linear acceleration (monitored through translational VOR responses) by measuring adaptive changes induced by exposure to a hyper-G environment. This specific aim addresses both Critical Questions (CQ) 9.01 and 9.04 by evaluating if Artificial Gravity combined with intermittent motion is a valid countermeasure (CQ 9.01) and by measuring adaptive changes induced by exposure to altered-gravity (CQ 9.04). We propose to measure human perception of orientation and eye movement responses during exposure to artificial gravity. To induce adaptation, we propose to utilize intermittent motion during centrifugation: passive roll tilt in a supine orientation will dynamically stimulate both semicircular canals and otolith organs, such that both sensory cues are correlated. (The motion envelope is in an Earth-horizontal plane.) For Specific Aim #1, we expect the human perceptual responses to be somewhat consistent with those previously measured in monkeys by Angelaki and Colleagues (1999), with the canal cue essential for accurate estimation of gravity and linear acceleration, but we also expect differences in eye responses as shown by our preliminary data with a significant horizontal eye response to roll tilt at high frequencies. The GIF resolution hypothesis states that the CNS estimates gravity and linear acceleration such that their mathematical difference matches the graviceptor measurement of the GIF. Therefore, for Specific Aim #2, based on the GIF resolution hypothesis, we expect that: a) the human responses for roll tilt about an earth-vertical axis in a supine orientation with a centripetal 1-G linear acceleration (Expt 2A) to be similar as for roll tilt about an earth-horizontal axis in an upright orientation in a 1-G environment since both dynamic canal and otolith cues are the exact same (except for a static naso-occipital 1-G gravitational acceleration in 2A): such a result would validate artificial gravity combined with intermittent motion as a countermeasure. The responses may not be exactly similar during Trial 1 but we expect the CNS to have adapted by Trial 4 back to a response similar to the response in Expt. 1 (1-G environment). b) the human responses for roll tilt about an earth-vertical axis in a supine orientation with a centripetal 1.2-G (or 1.4 G) linear acceleration (Expt 2B) to be originally (Trial 1) different from responses for roll tilt about an earth-horizontal axis in an upright orientation in a 1-G environment: for example, we expect a larger vertical VOR response (Trial 1). Then we expect the CNS to adapt to this Hyper-G environment: for example, the vertical VOR response should decrease while the horizontal VOR response increases as the CNS adapts to the hyper-G environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Since all proposed experiments are ground studies, the Earth Benefits should be straight forward. Experimental data shoud provide a better insight into how the Central Nervous System separates the graviceptor measurements of gravito-inertial force into estimates of gravity and linear acceleration, or in other words, into estimates of tilt and translation. This central process is essential to a number of everyday task such as visual stabilization or postural control and its understanding is therefore essential to the understanding of these many central processes. In addition, these ground studies should also provide insights on how the Central Nervous System adapts this separation process when humans experience a new gravitational environment. Understanding this process should not only help train and condition astronauts to microgravity or to a return to Earth-gravity after a long stay in microgravity. But it should also help us understand the rehabilitation process in vestibular patients and it should therefore help us design better methods of rehabilitation.
	The PI has completed the second year of this Task: it was principally dedicated to the design, delivery and installation of the new devices necessary to conduct the studies proposed in the PI's NASA grant # NNJ04HF79G. In addition, the PI started collecting experimental data for Expt. 1A. The Swing has been delivered at the beginning of June 2005. Set-up, tune-up and necessary modifications for the Swing have ended early March 2006 and we have started collecting experimental data for Expt. 1A two weeks ago. On the other hand, the Eccentric Rotator is now being assembled and scheduled for delivery in the 2nd quarter of 2006. 1. Devices: Swing - A new motion device, which is referred to as the "Swing" has been delivered in June 2005 and is fully operational as of March 15, 2006. This device has been purchased by Dr. Merfeld's NASA grant # NNJ04HB01G. This device is essentially a controllable swing that is able to provide tilts of up to 30°, angular velocities up to 200°/s, angular

accelerations up to 1200°/s/s. A short linear track has been also included to align the subject's naso-occipital axis with the roll tilt rotation axis. Both motions (tilt and vertical translation) are controlled via feedback servocontrol amplifiers and custom software that is presently in use in our lab.

Therefore, we have started collecting experimental data for Expt. 1A on this new Swing while our new eccentric rotator is currently being assembled. Expt. 1A was originally planned to be conducted on the tilting mechanism that is added to the eccentric rotator. But due to delays in Neurokinetics construction and delivery of the Eccentric Rotator, both PI and Co-I decided to conduct Expt. 1A on Dr. Merfeld's Swing to avoid losing any time.

Eccentric Rotator - We expect to receive our "Eccentric Rotator" in the second quarter of 2006\. This device is being purchased in part by the PI's NASA grant with other resources also contributing. The new eccentric rotator will be capable of creating three motion profiles: rotation, translation, and tilt. The main axis is a direct drive motor that rotates about an earth vertical axis. The translation axis moves the subject radially, relative to the yaw axis, at up to 2g combined acceleration. The subject will be tested in sitting and supine positions. The roll/tilt assembly will provide limited rotation (+/- 30 degrees) with the subject in either a seated position or in supine position, with head center on the tilt axis (the feet sweep an arc of 60 degrees). The system allows for independent rotation, translation (maximum radius of 70cm from center and up to 2G total head acceleration).

A Swing will be mounted on our eccentric rotator. The Swing will be able to provide tilts of up to 30° amplitude and the linear translator is capable of providing up to 2 G maximum. These motions will be combined to yield combined translation and tilt stimulation as originally proposed. As before, both motion axes (tilt and horizontal translation) will be controlled via feedback servocontrol amplifiers and custom software that is presently in use in our lab.

In the original grant application, the Eccentric Rotator was supposed to be delivered in 3 phases: tilt mechanism alone (upright orientation) in Year 1, eccentric rotator with tilt mechanism upright in Year 2, modification of tilt mechanism for supine orientation in Year 3. All designs have been completed. But due to the construction delays, phase 1 (tilt-mechanism alone) has been replaced by the use of the Swing to complete Expt 1A by the end of the 2nd quarter of 2006. Then, instead of delivering separately the tilting mechanism in the upright orientation and the eccentric rotator, both will be delivered simultaneously by the end of the second quarter of 2006.

2. Planned experiments:

All of the proposed investigations will be performed on human subjects, because the knowledge gained from testing humans more directly leads to countermeasure development. All of the proposed investigations will also be performed in darkness to minimize visual influences. As described in the original grant, we have chosen two behavioral measures for our studies: 1) eye movement recordings, and 2) psychophysical measures of motion perception.

Task Progress:

Experiment 1A: Roll Tilt - Eight subjects will be sinusoidally rotated in roll ±20° (0.342 G shear force) or ±10° (0.174 G shear force) about an axis of rotation near the center of the subject's head (Fig. 1). Subjects will be oscillated at 7 frequencies (0.01, 0.02, 0.05, 0.1, 0.2, 0.5 and 1.0 Hz) using the Swing. To further investigate high-frequency responses, we will rapidly (~ 0.3 s) roll tilt (5°, 10°, 15° and 30°) the subjects toward the left or right down orientation and hold them in this tilted orientation for 10 s (referred below to as "tilt steps").

Experiment 1B: Linear Translation (Linear Acceleration) - Eight subjects will be sinusoidally translated along their interaural axis (Fig. 1) with a peak linear acceleration of 0.342 G or 0.174 G, matching the shear force achieved in the roll tilt test paradigm (Experiment 1A). Subjects will be oscillated at the same 7 frequencies (0.01, 0.02, 0.05, 0.1, 0.2, 0.5 and 1.0 Hz) using the new eccentric rotator. The 0.5-1Hz trials will be achieved by simply translating along a 0.9-meter track. Because of the limited track length, the low frequencies 0.01-0.2 Hz trials will be conducted using the centrifuge to simulate a linear sled with a very long track length. Transient acceleration trapezoids will also be utilized (= "translation steps"). The acceleration trapezoid will be designed to precisely match the interaural shear force of Expt. 1A for about 0.5 s at 0.342 G, which is the maximum duration that we can accelerate the subjects given the limited track length.

Experiment 1C: Combined Tilt and Translation - The tilt (Expt. 1A) and translation (Expt. 1B) protocols described above will be performed simultaneously. The relative phase of the two sinusoidal stimuli (roll tilt and linear acceleration) will be varied. In one pairing the relative phase will result in the interaural shear force due to tilt approximately canceling the interaural shear force due to linear acceleration. In another pairing (180° different from the previous pairing), the two shear forces will augment one another, nearly doubling the interaural shear force measured by the otolith organs. Step stimuli will also be paired to yield no interaural force or to double the interaural force.

3. Results:

Since the Swing is now fully operational, we started testing subjects for Expt. 1A. Three subjects were sinusoidally rotated in roll ±20° (0.342 G shear force) about an axis of rotation near the center of the subject's head (see Pic. 4 in Appendix A1). Subjects were oscillated at 5 frequencies (0.02, 0.05, 0.1, 0.2, and 0.5 Hz). We simultaneously recorded eye movements and subjective roll tilt. Eye movement data are in agreement with previously published data, with amplitudes for both horizontal and torsional VOR responses increasing with increasing frequency. Amplitude of subjective roll tilts seems constant across the frequency range tested, with an increasing phase lead with increasing frequencies.

4. Future plans:

Since the Eccentric Rotator is currently being assembled, the PI feels very comfortable that Expt. 1A can be completed by the end of the third quarter of 2006, Expt. 1B by the end of 2006 and Expt. 1C by the end of the first quarter of 2007. Since funding ends in September 2006, a no-cost extension will be requested to conduct the proposed experiments beyond that date.

5. Updated Experimental schedule:

We deeply regret that NASA has decided to cancel funding of this grant beyond September 2006 while it initially was to end in May 2008. This funding decision practically suppresses support for half of the proposed experiments. Therefore, while the PI deeply regrets that Expts 2A-B cannot be investigated anymore with the revised allocated funds, the PI will request a no-cost extension to complete Expts 1A-C because of the delay in the Eccentric Rotator delivery. This no-cost extension has already been discussed with Mr. Dardano (BR&C Project Manager for Ground Based Research). We have therefore modified our experimental schedule accordingly.

Bibliography Type:	Description: (Last Updated: 12/02/2010)
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Articles in Peer-reviewed Journals	Zupan LH, Merfeld DM. "An internal model of head kinematics predicts the influence of head orientation on reflexive eye movements." J Neural Eng. 2005 Sep;2(3):S180-97. PMID: 16135883 , Sep-2005